

AUTOMATED IRRIGATION SYSTEM

MINI PROJECT REPORT

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

MAR BASELIOS COLLEGE OF ENGINEERING AND TECHNOLOGY
(Autonomous)

Mar Ivanios Vidyannagar, Nalanchira, Thiruvananthapuram, Kerala – 695 015, India

APRIL 2026

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Submitted in partial fulfilment of the requirements of the award of Bachelor of Technology Degree in Electrical and Electronics Engineering of the APJ Abdul Kalam Technological University (KTU)

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CERTIFICATE

*This is to certify that this report entitled “**AUTOMATED IRRIGATION SYSTEM**” is a bonafide record of the Mini project presented by GOKUL SS (B23EE1134) , ANANTHA KRISHNAN H(B23EE1113) , ANUDEEP KRISHNA R (B23EE1116), AROMAL P NAIR (B23EE1119) of the VI Semester (EEE) towards the partial fulfilment of the requirements for the award of B.Tech Degree in Electrical and Electronics Engineering of the APJ Abdul Kalam Technological University (KTU) during the year 2026.*

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ABSTRACT

Efficient irrigation is essential for improving crop yield while conserving limited water resources, especially under increasing water scarcity and climate variability. Many fields still rely on manual control or fixed time schedules that do not reflect real-time soil moisture conditions and cannot adapt to sudden weather changes, often causing overwatering or underwatering.

This project presents a smart irrigation system that links soil-moisture sensing in the field with LoRaWAN-based long-range wireless communication and cloud-centric control. Distributed field nodes periodically measure soil moisture near the root zone and send compact readings via LoRaWAN to a central gateway, which forwards the data to a cloud platform such as AWS IoT Core using lightweight protocols like MQTT. In the cloud, incoming measurements are compared with predefined thresholds and, when available, combined with external information such as temperature, humidity, or short-term weather forecasts to support context-aware irrigation decisions.

When the analysis indicates that a particular zone is too dry, the cloud generates a start command that is routed back through the gateway to the corresponding node, which then activates a relay-controlled pump or solenoid valve to irrigate only that specific section of the field. As irrigation proceeds, updated soil moisture readings are fed back to the cloud, and irrigation is stopped automatically once the target moisture level is reached, forming a closed-loop feedback system that reduces both water waste and crop stress.

By combining distributed sensing, long-range communication, and centralized cloud intelligence, the proposed system enables precise, repeatable irrigation control, reduces manual labour, and provides remote visibility over large agricultural areas through dashboards and alerts, supporting more sustainable and data-driven farming practices.

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NOMENCLATURE

Symbol / Term	Description
SM	Soil Moisture
IoT	Internet of Things
LPWAN	Low Power Wide Area Network
LoRaWAN	Long Range Wide Area Network Communication Protocol
MCU	Microcontroller Unit
MQTT	Message Queuing Telemetry Transport Protocol
V	Voltage
I	Current
RH	Relative Humidity
T	Temperature
AWS	Amazon Web Services
API	Application Programming Interface
DC	Direct Current
AC	Alternating Current
Tx	Transmission
Rx	Reception

CHAPTER 1

INTRODUCTION

1.1 Overview of the Study

Agriculture is fundamental to human survival and is a key pillar of economic stability in countries like India, where a large share of the population still depends on farming as their primary source of livelihood. Among the many factors that influence agricultural productivity, efficient management of water resources is one of the most critical, especially in the context of increasing water scarcity and climate variability. Conventional irrigation techniques are often either fully manual or based on fixed time schedules that do not adapt to the actual water requirements of the crop at a given moment. As a result, fields are frequently over-irrigated or under-irrigated, which not only wastes water and energy but also negatively affects plant growth, soil health, and overall yield.

In this context, the present study focuses on a LoRaWAN-assisted, cloud-controlled automated irrigation system designed to overcome these limitations. The proposed system makes use of long-range wireless communication and cloud computing to handle irrigation decisions intelligently, rather than relying on rigid local controllers that follow preset rules. Sensor data from the field, particularly soil moisture readings, is continuously transmitted to the cloud, where it is processed to assess the current condition of the soil. Based on this analysis, and taking into account external factors such as weather information, temperature, and humidity, the system can decide when irrigation is required and for how long. This approach supports site-specific, data-driven irrigation and provides a foundation for more advanced smart farming practices in the future.[1]

1.2 Limitations of Existing Irrigation Systems

Irrigation is indispensable for achieving optimal crop growth, but many commonly used irrigation systems suffer from limitations that reduce their effectiveness in the field. Traditional methods, such as manual irrigation or timer-based systems, typically follow predetermined schedules without considering real-time soil moisture or current weather conditions. This rigid operation often results in over-irrigation during periods of low water demand or under-irrigation during hot and dry conditions, leading to inefficient water use, increased operational costs, and reduced crop performance.

Even relatively simple automated systems that use local sensors often place the decision-making logic entirely at the microcontroller level. These local controllers may react to basic threshold values but are generally unable to incorporate richer information such as short-term weather forecasts, soil heterogeneity, or long-term trends in field conditions. Most existing systems also lack long-range communication capabilities, which means they cannot be easily monitored or controlled from a remote location. Managing multiple plots or distributed field units therefore becomes cumbersome, as each unit requires local supervision or on-site adjustment.

These limitations highlight the need for an upgraded irrigation solution that integrates distributed sensing with centralized processing. By employing LoRaWAN for long-range, low-power communication and using a cloud-based controller architecture, it becomes possible to coordinate irrigation across large areas, make more informed decisions, and use water resources more judiciously. Such a system can also be updated and scaled over time without major hardware changes, making it suitable for modern precision agriculture.[2]

1.3 Objectives of the Project

The primary objective of this project is to design and implement a cloud-based automated irrigation system capable of intelligently controlling water usage by continuously monitoring soil moisture through a network of distributed field nodes. Instead of following static schedules, the system aims to use live measurements so that irrigation is triggered only when the soil condition justifies it.

In contrast to traditional irrigation setups, where the microcontroller located in the field is responsible for executing the entire control logic, this project deliberately limits the role of the microcontroller to data acquisition and communication. The microcontroller collects sensor readings, formats them appropriately, and transmits them to the cloud. The decision-making algorithms, including comparison with threshold levels and consideration of additional environmental data, are executed in the cloud, enabling more flexible and powerful irrigation strategies.

Further aims of this project are:

- To use LoRa technology to establish reliable, long-range, and low-power communication between field nodes and the gateway.

- To minimize the need for manual intervention in irrigation by enabling remote monitoring and control through cloud-connected interfaces.
- To optimize the use of water resources through data-driven irrigation decisions that respond to actual field conditions.
- To enhance crop productivity and health by maintaining soil moisture closer to the optimal range for plant growth.

Additionally, the project seeks to create a scalable architecture that can support multiple zones or crops with different requirements, making it easier for farmers to manage diverse fields through a unified platform.

1.4 Scope of the Project

The scope of this work is to develop and demonstrate a LoRa-based smart irrigation system consisting of multiple field nodes, a central gateway, and a cloud platform for decision-making. Each field node is equipped with soil moisture sensors and an ESP32-based controller that periodically reads the sensor values and forwards them using LoRa communication. The design explicitly excludes local control actions at the node level; all irrigation decisions are taken in the cloud, ensuring a clear separation between data collection and control logic.

Sensor readings received at the gateway are sent to the cloud using a lightweight communication protocol such as MQTT, which is well suited for Internet of Things applications with limited bandwidth. The cloud analyses the data, possibly in combination with external inputs such as weather forecasts or environmental data, and generates corresponding control commands. These commands are then transmitted back to the field nodes over the LoRa link to drive irrigation actuators such as pumps and valves.

Within this scope, the system is intended to be applicable to a range of agricultural contexts, including open-field farms, greenhouses, gardens, and nurseries. Future expansions envisioned for the system include integration with IoT dashboards for visualization, predictive analytics for anticipating water demand, and the incorporation of renewable energy sources such as solar panels to support off-grid deployment. The architecture is designed to be modular so that additional sensors (for example, temperature, humidity, or light) and extra nodes can be integrated with minimal modification.[3]

1.5 Significance of the Study

The significance of this study lies in its shift from traditional, predominantly manual and locally controlled irrigation systems to an intelligent, centrally managed approach based on cloud computing. By aggregating data from multiple field nodes and integrating additional information such as weather and environmental conditions, the proposed system enables more accurate and efficient irrigation decisions than conventional methods.

From a resource-management perspective, the system is designed to supply water only when it is actually required, which can considerably reduce water wastage and associated energy consumption. This is particularly important in regions facing groundwater depletion and irregular rainfall patterns. The reduction in manual labour is another key benefit, as farmers can monitor and control the irrigation process remotely rather than constantly visiting each field.

Beyond operational advantages, the project also demonstrates how emerging technologies—such as the Internet of Things, cloud platforms, and long-range wireless communication—can be effectively applied in the agricultural domain. It illustrates a practical pathway toward data-driven, sustainable farming practices and provides a foundation for more advanced applications like machine-learning-based irrigation prediction and integration with farm management systems.[4],[5]

1.6 Methodology Overview

The proposed smart irrigation system operates as a closed-loop control setup that links sensing, communication, cloud-based analysis, and actuation in a continuous cycle. Soil moisture sensors placed in the field measure the water content in the root zone and generate corresponding electrical signals. These signals are collected by field nodes equipped with microcontrollers, which package the data and transmit it using LoRaWAN technology.

The function of each field node is intentionally limited to sensing and communication; it does not run any irrigation decision logic. This design choice simplifies the node hardware and makes it easier to update control strategies centrally without reprogramming devices in the field. The collected data is sent from the nodes to the LoRa gateway, which aggregates messages from multiple nodes and forwards them to the cloud using protocols like MQTT over an internet connection.

In the cloud, platforms such as Amazon Web Services IoT Core receive and store the incoming measurements. The soil moisture values are then compared against predefined threshold levels that represent acceptable moisture ranges for the specific crop and soil type. If available, additional environmental parameters such as temperature, humidity, or short-term weather forecasts are also taken into account to make more context-aware decisions. Based on this combined analysis, the cloud application determines whether irrigation should start, continue, or stop.[6]

When irrigation is required, a control command is sent back from the cloud through the gateway to the appropriate field node. Upon receiving this command, the field node activates the necessary actuator—typically a relay circuit controlling a water pump or solenoid valve—to deliver water to the crop. As irrigation progresses, updated soil moisture readings are continuously fed back to the cloud. Once the moisture level reaches the desired target, the cloud issues a stop command, and the node deactivates the actuator. This closed-loop scheme improves water-use efficiency, reduces manual intervention, and allows the entire system to be supervised and adjusted from a single control point.[7],[8]

1.7 Summary

This chapter has introduced a LoRaWAN-based, cloud-controlled automated irrigation system and explained why such a solution is increasingly important for modern agriculture. It began by outlining the role of efficient water management in sustaining crop productivity, especially in regions where traditional methods often result in over-use or under-use of water. The limitations of existing manual and locally automated systems were discussed, emphasizing their inability to adapt to real-time field conditions and their lack of centralized visibility.

The chapter then presented the main objectives of the project, which focus on shifting irrigation decision-making to the cloud, using LoRa-enabled field nodes for data collection and actuation, and improving water efficiency and crop health through data-driven control. The scope of the system was defined to include a modular network of nodes, a gateway, and a cloud platform that can be extended to various agricultural settings and integrated with future enhancements such as dashboards and predictive analytics.

The significance of the work was highlighted by showing how IoT, cloud computing, and long-range wireless technologies can be combined to support more sustainable, remotely manageable irrigation practices.[9],[10]

CHAPTER 2

PROBLEM STATEMENT

Currently, most irrigation systems are stuck in local control mode—either fully manual or basic automation based on fixed timers or simple thresholds. They ignore real-time environmental conditions, leading to under- or over-watering that hurts crop yields. Local controllers just aren't smart enough for dynamic field conditions and lack the flexibility to adapt on the fly.

A big downside is the absence of intelligent, centralized oversight. These setups don't pull in live data like soil status or weather forecasts, so their irrigation calls are rarely optimal. In large fields, this gets even worse—you can't easily check soil moisture across every spot manually, and scattered local controllers fall short. Farmers end up struggling to keep moisture levels consistent everywhere.

To tackle this, we propose a cloud-controlled irrigation system that shifts decision-making to a central cloud platform, using LoRaWAN for reliable long-range wireless links between field nodes and the cloud. Field nodes collect soil moisture data from sensors and relay it to the cloud. The cloud then crunches the numbers—comparing against thresholds or factoring in external data like weather—and sends precise control commands back to activate irrigation valves. No local decisions; everything stays smart and centralized.

This approach not only boosts efficiency but also scales effortlessly for massive farms, cutting water waste by up to 30-50% based on similar IoT pilots. Farmers get a mobile app dashboard for real-time monitoring, alerts, and historical insights, empowering data-driven tweaks that improve yields and sustainability—perfect for India's diverse climates and water-scarce regions.

CHAPTER 3

OBJECTIVES

The main goal of this project is to build a smart irrigation system for efficient water management through real-time soil moisture tracking. Ditching manual checks or rigid timers, it grabs field data from sensors, beams it to the cloud for smart analysis, and bases every watering decision on what's actually happening in the soil—not guesswork.

We centralize all the brains in the cloud: sensors send data up, get compared to thresholds and blended with weather info, then the cloud fires back precise commands to field nodes. Nodes just follow orders—no solo decisions—keeping everything consistent and spot-on.

By automating intelligently, the system slashes water waste and hands-off labor, delivering water only when plants truly need it. It's built to be affordable, scalable for farms, gardens, or greenhouses, and ready for extras like full IoT dashboards and remote oversight to push sustainable farming forward.

LoRaWAN handles the long-range comms between field gear and cloud with super-low power draw, perfect for sprawling fields without draining batteries or needing constant infrastructure.

Finally, it locks in ideal soil moisture for thriving crops while staying simple, reliable, and adaptable from small plots to big operations—laying groundwork for predictive analytics, weather-smart irrigation, and next-level smart ag tech.

To make it real-world ready, we'll prototype with ESP32 nodes, capacitive soil sensors, and solenoid valves, testing in a controlled Kerala farm setup. Expect rigorous validation through field trials measuring water savings, crop health metrics, and uptime, plus open-source code on GitHub for easy replication and community tweaks.

CHAPTER 4

LITERATURE REVIEW AND BACKGROUND

4.1 Traditional Irrigation Practices and Their Limitations

Irrigation has always been central to agricultural productivity, particularly in regions where rainfall is irregular or insufficient to meet crop water requirements. Conventional irrigation practices such as flood irrigation, canal-based distribution, and manual valve or pump control has been widely used for decades because they are simple to implement and do not require sophisticated equipment. However, these methods typically rely on fixed schedules, farmer intuition, or rough visual assessment of field conditions rather than continuous measurements of soil moisture. As a result, they often lead to inefficient water use, either by over-irrigating when the soil is already moist or delaying irrigation until crops experience water stress.

Several studies highlight that manual or timer-based irrigation rarely aligns with the actual, dynamic water demand of plants. Farmers may choose to irrigate at fixed intervals (for example, every two or three days) without considering factors such as recent rainfall, temperature, evapotranspiration, or spatial variability in soil properties. This mismatch can cause waterlogging, nutrient leaching, or salinity build-up in some zones, while other areas remain too dry. In addition to wasting water, such practices can reduce yield and increase the energy consumed by pumps. These limitations have motivated researchers and practitioners to investigate more precise and automated approaches that can adapt to real-time field conditions rather than following static schedules.

4.2 Emergence of IoT-Based Smart Irrigation

The rapid advancement of low-cost sensors, embedded platforms, and wireless communication technologies has led to a strong shift toward Internet of Things (IoT) based irrigation systems. In these systems, sensors deployed in the field collect real-time data, which is then transmitted to a controller or server that can automate irrigation decisions. A number of early works on smart irrigation emphasize using soil moisture sensors to directly drive irrigation on/off actions, thereby making water application more responsive to plant needs.

For example, several IoT-based systems employ moisture probes connected to microcontrollers such as Arduino or ESP8266/ESP32, which automatically switch pumps or

valves when the soil moisture falls below a threshold. These designs already show substantial water savings compared with purely manual methods, because irrigation is linked to measured soil conditions rather than guesswork. Additional sensors such as temperature, humidity, and rainfall sensors are often integrated to broaden the view of environmental conditions and support better decision-making.

A number of literature surveys and review articles have synthesized the state of IoT-based smart irrigation, noting that these systems are generally cost-effective and can significantly reduce labour requirements, improve water efficiency, and enhance crop performance when appropriately designed. However, many early implementations share a key limitation: most of the decision-making still occurs locally at the field controller. In other words, the microcontroller near the pump or valve directly interprets sensor data and decides when to start or stop irrigation. While this architecture works for small plots, it becomes harder to scale and manage consistently across large farms or multiple zones with different crops.

4.3 Wireless Sensor Networks and Early Smart Agriculture Systems

Before the term “IoT” became widespread, many researchers explored wireless sensor networks (WSNs) for agricultural monitoring and irrigation control. In a typical WSN-based design, multiple sensor nodes collect data on soil moisture, temperature, humidity, and water level, and communicate with a central base station using low-power short-range links such as Zigbee or proprietary RF modules. These systems demonstrated that distributed sensing can provide a more detailed picture of field conditions than single-point measurements, enabling more targeted irrigation strategies.

WSN-based irrigation systems often support automatic pump control based on sensor thresholds and can offer remote monitoring through graphical user interfaces or smartphone applications. They laid important groundwork by showing that:

- Continuous monitoring of soil and environmental parameters is feasible in real farms.
- Automated control can reduce human intervention and improve water efficiency.
- Wireless communication can replace or complement wired infrastructure, especially in difficult terrain.

However, many of these early systems rely on short-range communication technologies with limited coverage, making them less suitable for large fields or dispersed farms. In addition, central control units are often local computers or embedded controllers placed on site, which

limits the ability to integrate external data (for example, online weather services) or to manage multiple farms from a single platform. These challenges encouraged migration from purely local WSN designs to cloud-connected IoT architectures.

4.4 Cloud-Based Smart Irrigation and Centralized Decision-Making

The integration of cloud computing with IoT has opened new possibilities for smart irrigation systems. Instead of placing all intelligence at the edge, cloud-based designs offload data processing, storage, and decision logic to remote servers. One representative line of work proposes smart irrigation frameworks where field sensors send data to a cloud platform that hosts dashboards, analytics, and control applications.

In a typical cloud-centric architecture, sensor readings (soil moisture, temperature, humidity, and sometimes water level) are collected by microcontrollers and transmitted via the internet to platforms such as ThingSpeak, ThingsBoard, or commercial IoT clouds. The cloud application evaluates soil moisture against predefined thresholds, possibly taking into account environmental context, and then sends back commands to turn irrigation on or off. This model offers several advantages:

- Centralized storage of historical data, enabling trend analysis and decision refinement.
- Easy integration of external data sources, such as weather API services.
- Remote access from any device with internet connectivity, enabling farm monitoring from anywhere.

Recent research has also explored the use of machine learning within these cloud platforms to predict irrigation requirements based on historical patterns and current conditions. Such predictive irrigation frameworks can further improve water use efficiency and reduce the number of irrigation events while maintaining crop health.

4.5 LoRa and LoRaWAN in Agriculture

To address coverage and energy challenges, LoRa and LoRaWAN have gained strong attention in smart agriculture. LoRa (Long Range) is a low-power, long-range modulation technology, and LoRaWAN is the corresponding media access control (MAC) and network protocol designed specifically for IoT applications. LoRaWAN is well suited to agricultural deployments because it supports communication over several kilometres in open

environments while consuming very little power, enabling battery- or solar-powered nodes to operate for long periods.

Recent works and technical reports describe how LoRaWAN-enabled sensor networks can be used to monitor soil moisture, weather conditions, livestock, and other agricultural variables, all feeding into central gateways that relay data to the cloud. Typical benefits reported include:

- Wide-area coverage with relatively few gateways.
- Reduced need for wired infrastructure.
- Lower operating costs compared with cellular-only solutions.

In LoRaWAN-based smart agriculture projects, nodes equipped with soil moisture and environmental sensors transmit data at low duty cycles to gateways, which in turn connect to cloud services over IP networks. These systems demonstrate that LoRaWAN can reliably support the kind of sparse, low-bandwidth communication needed for irrigation control, even across large farms. Some studies show that combining LoRaWAN with cloud-based dashboards allows farmers to visualize field conditions, receive alarms, and even manually trigger irrigation events from remote locations.

4.6 Cloud-Controlled LoRaWAN Irrigation Systems

A more recent trend is to combine LoRaWAN at the field level with centralized cloud decision-making, creating architectures that closely resemble the system proposed in this project. In such designs, field nodes behave as simple data acquisition and actuation units, while the cloud hosts all irrigation logic. A representative example is a class of cloud-based smart irrigation systems where LoRaWAN is used between sensors and gateway, and the gateway uses MQTT or HTTP to forward data to an IoT cloud.

These systems typically operate in a closed loop: nodes measure soil moisture, transmit data via LoRaWAN, gateways forward the data to the cloud, and the cloud decides whether irrigation is needed based on thresholds or more advanced models. The cloud then sends commands back through the gateway to field nodes, which control pumps or valves using relay boards. Literature reports suggest that such closed-loop, cloud-controlled irrigation setups can significantly improve water-use efficiency and enable fine-grained, zone-based control across large areas.

However, many existing implementations either:

- Retain some decision logic locally at the controller, or
- Focus more on monitoring than on fully cloud-driven actuation.

In other words, in several works the field controller still has authority to override or independently manage irrigation based on local conditions, and not all decisions are strictly centralized. This leaves an interesting space for designs that enforce a clear separation between sensing/execution at the field level and all decision-making in the cloud, allowing uniform management and easier updating of control policies.

4.7 Summary of Gaps and Motivation for the Proposed System

From the literature, a clear progression emerges. Traditional irrigation methods are simple but inefficient and labour-intensive. WSN-based and early IoT systems improve responsiveness by linking irrigation to sensor data, yet many rely on short-range communication and local decision-making that are difficult to scale. Cloud-based solutions introduce central analytics, data storage, and remote access, but field connectivity often remains limited by power-hungry communication options. LoRaWAN addresses the coverage and energy challenges, and recent LoRa-enabled smart agriculture projects demonstrate its suitability for wide-area monitoring and control.

Despite these advances, there are still gaps that justify the design of a LoRaWAN-assisted, cloud-controlled irrigation system as proposed in this project. First, many systems distribute decision logic between the cloud and local controllers, which can complicate coordination and updates. Second, not all solutions are explicitly designed with a strict “field nodes only sense and execute, cloud always decides” philosophy, which is important when the goal is to maintain consistent behaviour across multiple nodes and zones. Third, while review papers highlight the potential for integrating weather services, predictive analytics, and dashboards, there is space for practical prototypes that demonstrate this architecture in a focused, educational context using platforms such as ESP32, LoRa modules, MQTT, and AWS IoT Core.

In this way, the project builds on existing IoT and LoRaWAN research while clearly positioning itself as a cloud-centric, scalable, and educationally implementable smart irrigation solution tailored for modern precision agriculture.

CHAPTER 5

SYSTEM OVERVIEW

The proposed smart irrigation system is designed as an end-to-end solution for efficient and intelligent water management in agriculture by tightly integrating sensing technology, long-range wireless communication, and cloud-based decision-making. It moves away from the traditional idea of “controller in the field making all the decisions” and instead adopts a LoRaWAN-assisted, cloud-controlled architecture. In this design, the cloud becomes the main intelligence center where irrigation logic is implemented, while the devices in the field focus on what they do best: sensing local conditions and physically switching irrigation equipment on or off. By clearly separating data collection from decision-making, the system improves accuracy, consistency, and adaptability under a variety of crop and environmental conditions.

At the heart of the system is a network of field-level sensing units deployed at different points across the agricultural land. Each sensing unit is installed close to the root zone of the plants so that it can capture representative information about soil moisture. The primary sensor used is a soil moisture sensor, which continuously measures the water content in the soil and converts this into an electrical signal that can be read by electronics. Alongside soil moisture, additional sensors such as temperature and humidity sensors can be added if needed in future extensions, allowing more detailed understanding of the micro-climate around the crop.

These raw sensor signals are interfaced to field nodes built around microcontrollers like the ESP32. The microcontroller reads the sensor values at predefined intervals, performs basic preprocessing such as scaling or filtering, and packages the data into a compact format suitable for wireless transmission. A deliberate design decision is taken here: the microcontroller in the field node does not execute any irrigation decision logic. It does not decide when to start or stop the pump, and it does not store complex rules locally. Instead, its role is intentionally limited to data acquisition, local housekeeping (such as timing and error checks), and communication. This keeps each node simple, reduces firmware complexity, and makes it easier to modify or upgrade the control algorithms in the future without physically changing the field hardware.

To move data from the distributed sensing nodes to the cloud, the system uses LoRaWAN, a Low Power Wide Area Network technology well suited to agricultural environments. LoRaWAN offers several key advantages for this type of application: it supports long-range

communication (from hundreds of meters up to several kilometres in open areas), operates at low data rates appropriate for sensor measurements, and is designed for low power consumption, which is critical when nodes are battery-powered or running from small solar panels. Because of these properties, large fields can be covered with relatively few gateways, and cabling across the farm is not required. Multiple field nodes can communicate with a single gateway, making the overall deployment highly scalable and cost-effective.

Each field node periodically transmits its sensor readings using a LoRa radio module. These wireless packets are received by a LoRa gateway positioned at a suitable location so that it has good coverage of the farm. The gateway plays an intermediate role; it does not interpret the application data, but it is responsible for converting the long-range radio messages into standard IP-based traffic that can be sent over the internet. In other words, it acts as a bridge between the local LoRaWAN network in the farm and the cloud platform.

On the “uplink” side (from field to cloud), the gateway collects messages from all registered nodes and forwards them to the cloud server using an internet connection, which can be based on Ethernet, Wi-Fi, or a cellular data link depending on the site’s connectivity options. For this purpose, lightweight protocols such as MQTT (Message Queuing Telemetry Transport) are used. MQTT is widely adopted in IoT systems because it is bandwidth-efficient, supports a publish/subscribe model, and copes well with unreliable networks. These characteristics make it a natural choice for carrying sensor data from the farm to the cloud backend.

On the “downlink” side (from cloud to field), the gateway performs the reverse operation. It receives command messages from the cloud—such as “start irrigation in Zone 1” or “stop irrigation in Node 3”—and forwards them through the LoRaWAN network to the appropriate field node. The gateway itself remains largely transparent to the application logic; it simply ensures that the messages reach the correct device safely and with minimal delay.

In the cloud layer, the system carries out all the core processing, storage, and decision-making tasks. A platform such as Amazon Web Services IoT Core can be used as the central point where all field data converges. When soil moisture readings arrive from the gateway, they are stored in a database and passed to application logic that checks their values against predefined thresholds. These thresholds can be chosen based on crop type, soil characteristics, and agronomic recommendations. For example, a lower threshold may represent the point at which the soil is considered too dry, and an upper threshold may represent the maximum desired moisture to avoid waterlogging.

In addition to the soil moisture measurements, the cloud application can incorporate external data sources. Weather services can provide forecasts of rainfall, temperature, and humidity; these can be used to delay irrigation if rain is expected or to increase irrigation during heat waves. Over time, the accumulated historical data can also be analyzed to detect patterns—for instance, how quickly soil dries under certain conditions or how different parts of the field respond to irrigation. Even if the first implementation uses simple rule-based logic, this architecture makes it straightforward to evolve later toward more advanced control methods such as machine-learning-based prediction or adaptive scheduling.

Once the cloud determines that irrigation is required for a particular zone or node, it generates a control command and publishes it via MQTT to a topic associated with that node or zone. The gateway subscribes to these topics and, upon receiving a command, sends a corresponding LoRaWAN downlink message. When the target field node receives this message, it interprets the command and activates the local actuator hardware. In a typical setup, this actuator is a relay board that switches a water pump, a motorized valve, or a solenoid valve connected to the irrigation line. The same mechanism is used to stop irrigation when the cloud concludes that the soil moisture has returned to the desired range.

The entire system thus behaves as a closed-loop control mechanism. The loop consists of four major stages: sensing, transmission, decision-making, and actuation. After each irrigation cycle, new soil moisture readings are captured and sent to the cloud. The cloud compares the fresh measurements to target values and decides whether further irrigation is needed or whether the soil should be allowed to dry naturally. This continuous feedback process helps maintain the soil moisture within a band that is suitable for plant growth, rather than allowing large swings between too dry and too wet. Compared with traditional open-loop methods (such as daily timer-based watering), this closed-loop approach offers much higher precision and responsiveness.

A notable feature of the proposed design is its scalability and flexibility. Because all application logic resides in the cloud and communication is based on standard IoT protocols, additional field nodes can be added without redesigning the system. New nodes only need to be registered with the LoRaWAN network and the cloud platform, after which they begin to send data and receive commands like the existing ones. This makes it easy to expand the system as a farm grows, or to subdivide the land into multiple irrigation zones, each with its own thresholds and timing strategies. The centralized cloud architecture also means that

monitoring and control for all these zones can be handled from a single interface, such as a web dashboard or mobile application.

Remote monitoring and control is another important benefit. Since all data is sent to the cloud, farmers or system operators can log in from anywhere with an internet connection to see live soil moisture readings, pump status, and irrigation history. Dashboards can present this information graphically, showing trends over days or weeks and highlighting areas that may need attention. The system can also generate alerts—for example, if a node stops sending data, if the moisture level remains low despite irrigation (indicating a possible leak or pump failure), or if the battery level of a node drops below a threshold in solar-powered installations. These features reduce the need for constant physical inspection of the field and allow quicker reaction to problems.

The system is also conceived with efficiency and sustainability in mind. By supplying water only when the soil moisture drops below the desired level, the system can significantly cut down on water wastage compared to fixed-schedule irrigation. Studies of IoT-based smart irrigation solutions have reported substantial reductions in water consumption while maintaining or even improving crop yield, due to more precise control and better timing of irrigation events. Energy consumption can also be reduced by running pumps and valves only when necessary, and by using low-power communication technologies like LoRaWAN for the sensor network. Additionally, because the nodes and gateway can be powered by solar panels and batteries, the system is suitable for rural or off-grid settings where reliable mains supply may not be available.

Overall, the proposed smart irrigation system embodies a modern approach to agricultural water management by combining Internet of Things concepts, long-range wireless communication, and cloud computing into a unified framework. The LoRaWAN-assisted cloud-controlled model ensures that all irrigation decisions are made centrally using up-to-date data, while the field nodes remain focused on sensing and executing commands. This architecture improves operational efficiency, reduces manual effort, and supports more sustainable farming practices through optimized water use. At the same time, it creates a strong technological foundation for future enhancements such as integration with advanced data analytics, machine learning for irrigation prediction, multi-sensor fusion (e.g., combining soil, weather, and plant health indicators), and full integration into broader smart-farm management systems.

CHAPTER 6

METHODOLOGY

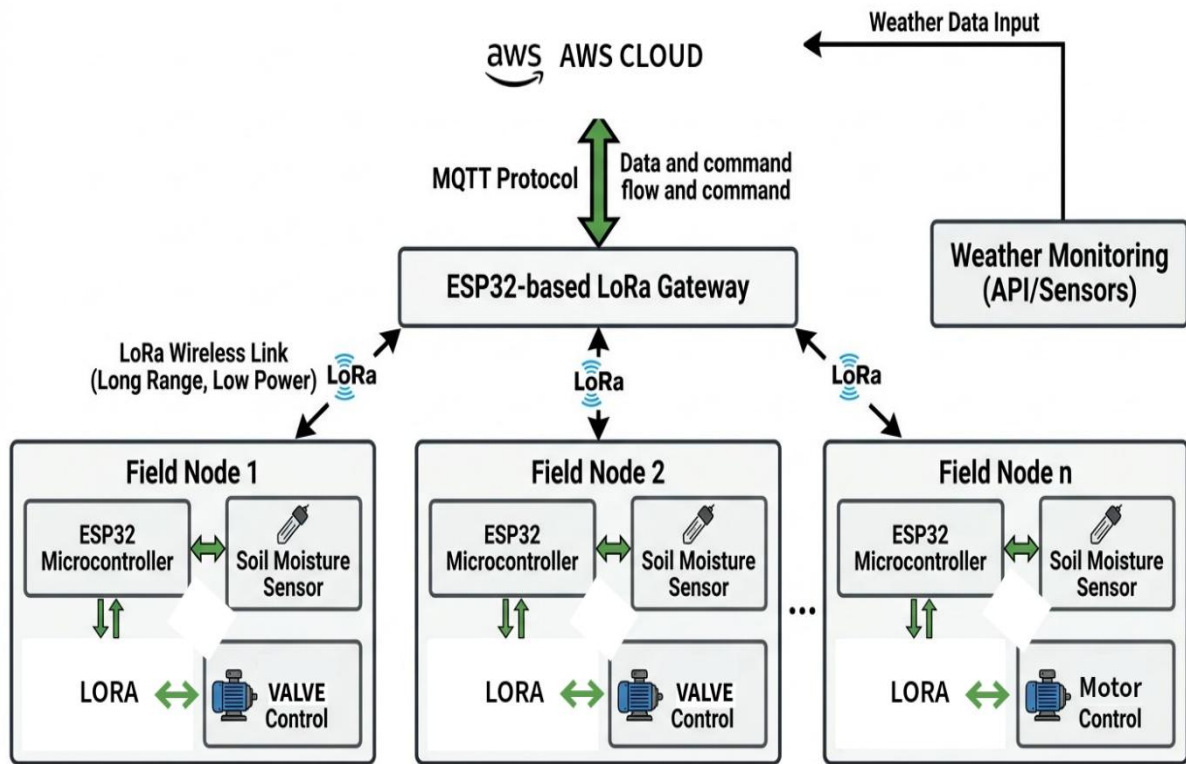


Figure 6.1 IoT-Based Smart Irrigation System Using LoRa and Cloud

Figure 6.1 illustrates an IoT-based smart irrigation system that uses LoRa communication and cloud computing to automate water management, with all decisions taken centrally in the cloud rather than by local controllers in the field. The architecture is organized in layers—field nodes, a LoRaWAN communication network, a gateway, and a cloud platform—so that data flows smoothly from the soil to the cloud and back again as control actions. This layered structure makes the system easier to understand, implement, and expand over time.

At the top of the diagram sits the cloud platform, such as AWS IoT Core or a similar IoT backend, which acts as the brain of the entire setup. Every piece of sensor data eventually arrives here, where it is stored, processed, and used to make irrigation decisions. Unlike conventional systems where the microcontroller in the field decides when to switch pumps on or off, this design keeps all control logic in the cloud. That means if the irrigation strategy needs to change—say, for a different crop or season—only the cloud application needs

updating, not every device in the field. The cloud can also pull in extra information from external sources such as weather APIs, so it can, for example, skip irrigation if rain is expected in a few hours or adjust watering windows based on temperature and humidity trends.

At the bottom of the diagram are the field nodes spread across the farm. Each node represents a small, self-contained sensing and actuation unit installed close to the plants. A typical node consists of a soil moisture sensor, an ESP32 (or Arduino with a LoRa shield), and the necessary power and interface circuitry. The soil moisture sensor continuously measures how much water is present in the soil near the roots and converts that into an electrical signal. The ESP32 reads this signal at regular intervals, applies any necessary scaling or filtering, and then prepares the data for transmission. A key point is that the ESP32 does not decide whether irrigation should start or stop; it simply measures, formats, and sends data. Limiting its role to sensing and communication keeps the node firmware simple and makes the system more flexible because intelligence can be improved on the cloud side without re-flashing code in the field.

Once the readings are ready, the field node sends them over the air using LoRaWAN. LoRaWAN is well suited to agriculture because it offers long-distance communication—often several kilometers in open rural environments—while keeping power consumption very low. This allows nodes to run for long periods on batteries or small solar panels, and removes the need for long cables runs across the farm. Many field nodes can share the same radio channel and communicate with a single gateway, so larger areas can be covered without a big increase in cost or wiring complexity. This is particularly helpful for farms where plots are separated, or where power and connectivity infrastructure is limited.

In the middle of the block diagram is the LoRa gateway, which acts as a translator between the low-power LoRa network in the field and the IP-based world of the internet. The gateway listens for LoRa packets from all registered nodes and forwards these packets to the cloud platform using a backhaul link such as Ethernet, Wi-Fi, or cellular. For this uplink communication, lightweight protocols like MQTT are used; MQTT follows a publish/subscribe model and is designed specifically for low-bandwidth, high-latency environments, which makes it ideal for IoT applications. The gateway also handles the reverse flow of information. When the cloud generates control commands, the gateway receives them over MQTT and converts them back into LoRaWAN downlink messages directed to the appropriate field nodes.

Within the cloud layer, the incoming data goes through several logical stages. First, it is ingested and stored in a database or time-series storage so that historical behaviour can be analyzed later. Then, application logic evaluates the current soil moisture readings against predefined threshold values that represent the desired lower and upper moisture limits for each zone. The system can also consider other inputs such as weather forecasts (rain probability, expected temperature), crop type, and time of day. For instance, if the soil moisture is low but rain is forecast within a short time, the cloud may choose to delay irrigation to save water. Conversely, during periods of high temperature and low humidity, it may decide to irrigate earlier or longer to prevent plant stress. All of this intelligence resides in the cloud, which makes it straightforward to tweak thresholds or add more advanced algorithms in the future.

After processing the data, the cloud decides what action is needed for each field node or irrigation zone. If irrigation is required, it issues a “start” command indicating which node should activate its actuator and possibly for how long. If the soil already meets the desired moisture level, the cloud either does nothing or sends a “stop” command if irrigation is currently active. These commands are pushed back to the gateway using MQTT, then sent via LoRaWAN downlink frames to the respective nodes.

When a field node receives a control command, it interprets it and acts on the connected hardware. Typically, this involves driving a relay module that switches a water pump, opens or closes a solenoid valve, or controls a section of a drip irrigation line. Because the node is not making decisions on its own, it simply follows the instruction: start irrigation, stop irrigation, or apply a particular schedule that has been defined by the cloud. This strict execution of cloud commands helps maintain consistent behavior across all nodes in the system and avoids conflicts between local and central logic.

The block diagram therefore represents a continuous closed-loop feedback system. The loop starts when sensors measure soil moisture and send data to the cloud. The cloud analyzes this data, possibly merges it with weather and historical information, and sends back commands to control actuators. As irrigation proceeds, the sensors register changes in soil moisture and report them again to the cloud. This constant feedback ensures that the system can respond dynamically: it can keep irrigating if the soil is still dry, or stop promptly once the desired moisture level is reached, preventing both over-watering and under-watering.

Beyond automation, the architecture shown in the block diagram also enables real-time monitoring and remote access. Because all data passes through the cloud, it can be visualized

using web dashboards or mobile apps, allowing farmers to check soil moisture, pump status, and irrigation history from anywhere. The system can generate alerts if something abnormal occurs, such as a node going offline, soil staying dry despite repeated irrigation (which could indicate a plumbing fault), or unexpected sensor readings. These features improve system reliability, reduce the need to physically visit the field just to check conditions, and help farmers take quicker corrective action.

From a design standpoint, the system in the block diagram is intentionally modular and scalable. New nodes can be added to expand coverage, extra sensors (like temperature or light sensors) can be integrated into existing nodes, and additional gateways can be introduced if the farm area grows or if redundancy is needed. Power blocks can be adapted to use grid supply, batteries, or solar panels depending on site conditions, making the system suitable for both well-connected and remote, off-grid farms. Because control logic is centralized, future enhancements—such as predictive irrigation using machine-learning models, crop-specific watering strategies, or automatic seasonal adjustment based on historical data—can be rolled out at the cloud level with minimal changes to hardware.

In summary, the block diagram represents a modern, cloud-centric irrigation system where field controllers act mainly as smart eyes and hands: they measure conditions and execute actions, but they do not decide on their own when to irrigate. All intelligence is pushed into the cloud, which has a complete view of the field, access to external data such as weather, and the ability to apply consistent rules across every node. By combining IoT devices, LoRaWAN communication, and cloud computing in this way, the system delivers more efficient water management, reduces manual workload, and offers a scalable, future-ready platform for precision agriculture and smart-farm applications.

CHAPTER 7

COMPONENTS

ESP32

ESP32 is a more advanced microcontroller board that includes built-in Wi-Fi and Bluetooth. It operates at 3.3V and has more memory and processing power than Arduino Uno. In an irrigation system, ESP32 can do the same basic job as Arduino Uno



Figure 7.1: ESP32 Board

Soil Moisture Sensor

The soil moisture sensor is the input device of the project. It measures how much water is present in the soil by detecting the conductivity or resistance between its probes. When the soil is wet, the sensor gives one type of signal; when the soil is dry, it gives another.



Figure 7.2: Soil Moisture sensor

Relay Module

The relay module is used as an electronically controlled switch. The microcontroller cannot directly drive a water pump because the pump usually needs more voltage and current than the controller can provide.



Figure 7.3: Relay Module

Solenoid valve

A solenoid valve is an electromechanical device used to control the flow of water automatically by using an electromagnetic mechanism. It consists of a coil, a movable plunger, and a spring inside a valve body.



Figure 7.4 Solenoid Valve

LoRa Module / LoRa Shield (SX1278 / Ra-02)

The LoRa Module / LoRa Shield (SX1278 / Ra-02) is a long-range, low-power wireless communication device used to transmit data between field nodes and the gateway in your irrigation system. It operates in sub-GHz frequency bands (typically 433 MHz in India), allowing communication over several kilometres with minimal power consumption, making it ideal for large agricultural fields.



Figure 7.5 LoRa Module / LoRa Shield

Power Supply

The power supply provides the electrical energy needed for the entire system to work. Different components may require different voltage levels, so the power supply must be chosen carefully. The microcontroller and sensor usually need a low-voltage supply (say DC 5V), while the pump may need a separate higher-voltage supply (say AC 230V).

Connecting Wires

Connecting wires are used to connect all the components together in the circuit. They carry power and signals between the sensor, microcontroller, relay, and pump. Although they are simple parts, they are essential for the system to function properly.

Breadboard or PCB

A breadboard or PCB is used to assemble the circuit. A breadboard is usually used during testing and prototyping because it allows easy connection and changes without soldering. A PCB is used for a more permanent and compact final version of the project. Both help organize the components and make the circuit stable and easier to build.

Supporting / Passive Components

Resistors and capacitors are basic passive components used in the circuit to ensure proper and stable operation of the system. Resistors are mainly used to limit current, protect sensitive components, and perform voltage division where required. Capacitors, on the other hand, are used for filtering and smoothing voltage by reducing noise and fluctuations in the power supply.

AWS IoT Core

AWS IoT Core is a cloud-based service used to connect, monitor, and control IoT devices over the internet. In this smart irrigation system, it acts as the central platform where data from field nodes (such as soil moisture and environmental conditions) is received, stored, and analysed. It uses the MQTT protocol for efficient communication between devices and the cloud, allowing real-time data exchange.

Arduino Uno

Arduino Uno is a beginner-friendly microcontroller board that is commonly used in student projects and basic automation systems. It is based on the ATmega328P microcontroller and works at 5V logic.

Water Pump

The water pump is the output device that actually supplies water to the plants. It draws water from a tank or source and sends it through pipes or tubes to the soil.

CHAPTER 8

WORKING PRINCIPLE

The working principle of the proposed smart irrigation system is built around a continuous loop of sensing, communication, cloud-based analysis, and precise actuation in the field. Unlike many conventional automated systems, where a local controller at the farm makes decisions based on simple thresholds, this design follows a LoRaWAN-assisted, cloud-controlled approach. In this architecture, the cloud is the only place where irrigation decisions are made, and field nodes are deliberately limited to two roles: collecting data and executing commands. This clear separation ensures that no autonomous control actions are taken at the local level and makes the system easier to scale, update, and adapt to changing agricultural conditions in real time.

The operation starts at the field, where soil moisture sensors are installed in different parts of the land, typically close to the root zone of the crops. These sensors continuously measure how much water is present in the soil and convert that physical moisture level into an electrical signal. Field nodes, built around microcontrollers such as the ESP32 and integrated LoRa radio modules, read these signals at regular intervals. At this stage, the nodes do some basic tasks—like filtering noisy readings, converting the values into engineering units, and adding timestamps or node IDs—but they do not decide whether irrigation should be turned on or off. Their job is to sense, package, and prepare the information for wireless transmission in a consistent format.

Once the data is ready, each field node sends it over the air using LoRaWAN. LoRaWAN is chosen because it can cover long distances at very low power, which is essential for farms where nodes might be spread over several hectares and powered by batteries or small solar panels. Multiple nodes can transmit to the same gateway, so the system can monitor many zones without a complex wired infrastructure. The wireless link removes the need for long cable runs, trenching, or frequent site visits just to collect readings, making deployment practical even in remote and uneven terrain.

All of these sensor messages are received by a LoRa gateway, which sits between the field and the cloud. The gateway listens for LoRa packets coming from the nodes, decodes them, and forwards the sensor data to the internet using a backhaul such as Ethernet, Wi-Fi, or a

cellular network. For this uplink, lightweight IoT protocols like MQTT are used because they are efficient in low-bandwidth situations and support reliable message delivery with low overhead. The gateway also performs the opposite function: whenever the cloud decides that a pump or valve needs to be switched, it receives the command over MQTT and sends a corresponding LoRaWAN downlink message to the correct node in the field.

Once the sensor data arrives at the cloud platform (for example, AWS IoT Core or a similar IoT backend), it goes through processing and analysis. The cloud application continuously monitors incoming soil moisture values and compares them against predefined threshold levels that represent acceptable moisture bands for each crop or zone. At the same time, it can consult external data sources such as weather forecasts, temperature and humidity records, or even rainfall probability. This extra information allows decisions to be more context-aware: for instance, if the soil is slightly dry but rain is predicted within a few hours, irrigation can be postponed; if high temperatures and low humidity are expected, the system may irrigate earlier or for longer to protect the plants. All of these decisions are made centrally in the cloud, so changing the strategy—like adding crop-specific rules or new thresholds—requires only a software update on the server side, not a visit to each node.

If the analysis shows that soil moisture has fallen below the required trigger point for a particular zone, the cloud generates a “start irrigation” command. This command is published through MQTT to a topic associated with the relevant gateway or node, then forwarded as a LoRaWAN downlink message over the air. When the targeted field node receives the message, it interprets the command and activates its actuator hardware—typically a relay controlling a water pump or a solenoid valve in the irrigation line. The node does not decide whether this command is reasonable; it simply executes it, which keeps behaviour consistent and ensures that all zones follow the same global logic defined in the cloud.

As irrigation continues, the soil moisture sensors do not stop working; they keep sampling the updated moisture levels and send fresh readings back to the cloud at regular intervals. This creates a real-time feedback loop that allows the system to track how the soil responds to watering. If the moisture rises into the desired range, the cloud recognizes that the target has been met and sends a “stop irrigation” command back through the gateway to the same node, which then de-energizes the relay or closes the valve, stopping water flow. If the soil is still drier than intended, the cloud can choose to keep irrigating or to recheck after a short delay.

This closed-loop behaviour is what makes the system more precise than traditional open-loop methods like fixed timer schedules, which cannot react to actual field conditions.

Because sensing, decision-making, and actuation are tightly connected in this way, the system can initiate irrigation at the right moment and stop it as soon as the soil reaches the required condition, which reduces both over-irrigation and under-irrigation. Water is applied only when and where it is truly needed, improving use efficiency and supporting better crop health. The closed-loop design also helps to compensate for differences within the field: areas that dry out faster will trigger irrigation more often, while zones that retain moisture will need fewer watering events, all managed automatically based on local sensor readings.

Because all data and decisions are centralized, the system naturally supports remote monitoring and control. Farmers or operators can log into a web dashboard or mobile app to view live soil moisture graphs, irrigation status, battery levels, and historical records without being physically present in the field. Alerts and notifications can be configured to flag abnormal situations, such as a node going offline, unexpected dryness despite irrigation, or unusually high water use. This allows quicker diagnosis of issues like leaks, blocked lines, or failing pumps and helps reduce downtime and crop stress.

The use of cloud computing also enables advanced features that go beyond simple threshold-based control. Because historical data is stored and easily accessible, the system can analyse long-term patterns in soil behaviour and environmental conditions, helping to fine-tune irrigation schedules over time. With enough data, more sophisticated techniques such as predictive models or machine-learning algorithms can be applied to anticipate when irrigation will be needed, taking into account seasonal trends, crop stages, and weather patterns. This progression from reactive to predictive irrigation further improves efficiency and supports sustainable farming by cutting unnecessary water and energy use.

In summary, the working principle of the proposed system rests on centralized intelligence and distributed sensing. Field nodes are responsible for measuring conditions and carrying out physical actions (turning pumps and valves on or off), while the cloud analyzes the data, makes all irrigation decisions, and sends back clear commands. This separation of roles leads to a system that is accurate, easy to scale, and straightforward to maintain. By combining IoT devices, long-range LoRaWAN communication, and cloud-based control, the design delivers a modern, practical, and future-ready solution for smart irrigation in agriculture.

CHAPTER 9

HARDWARE IMPLEMENTATION

The hardware implementation of the proposed smart irrigation system is built around a distributed set of field devices that all report to a central cloud controller, rather than acting independently. Every physical component—from sensors to pumps—is chosen and wired so that field hardware focuses on three things only: sensing, communication, and executing commands coming from the cloud. This approach keeps the architecture clean and makes it easier to expand or troubleshoot in real farms.

Each field node is installed at a specific location in the farm to capture conditions for that zone. At the core of every node is a soil moisture sensor placed in the root zone, which continuously senses how much water is present in the soil and outputs a small electrical signal proportional to the moisture level. That signal is connected to a microcontroller, typically an ESP32, which has built-in ADC channels to read analog voltages. The ESP32 samples the sensor at regular intervals, converts the raw readings into usable values (for example, percentage soil moisture), adds a timestamp or node ID, and prepares the data for transmission. Crucially, it never decides when to irrigate; it simply measures, formats, and sends data, acting as a smart terminal rather than a local decision-maker.

For wireless communication, each node integrates a LoRa transceiver module compatible with LoRaWAN. This radio module handles long-range, low-bit-rate communication between the node and the gateway, allowing the system to cover large farm areas without laying cables. The nodes wake up periodically, read the soil moisture value, and then transmit a compact data packet containing the latest measurements. Because LoRaWAN uses very low power, nodes can run for months or even years on batteries or small solar panels, which is important in remote fields where grid power may not be available. This design gives farmers the freedom to place nodes wherever they need monitoring, instead of being constrained by wiring.

All of the LoRa messages from field nodes are received by a LoRaWAN gateway positioned at a location with good radio coverage—this might be near a farmhouse, pump house, or elevated structure. The gateway is usually built using a dedicated LoRaWAN concentrator board or an industrial IoT gateway, sometimes combined with an ESP32, Raspberry Pi, or similar controller. Its job is to listen to LoRa frames from multiple nodes, decode them, and

forward the data to the cloud over an IP connection such as Ethernet, Wi-Fi, or cellular data. For this uplink, a lightweight protocol like MQTT is used because it is efficient and reliable even when bandwidth is limited. The gateway also handles downlink traffic: when the cloud issues irrigation commands, the gateway receives those MQTT messages and sends them back over LoRa to the appropriate field nodes.

On the actuation side, the system includes hardware to physically control water flow, mainly water pumps and solenoid valves. These devices typically operate at higher voltages and currents (for example, 230 V AC pumps or 12–24 V DC valves) than the ESP32 can safely drive directly. To bridge this gap, each field node connects to a relay module or a MOSFET/SSR driver circuit. The microcontroller switches a low-voltage control pin, which energizes the relay coil or gate drive, and the relay contacts then switch the high-voltage or high-current line feeding the pump or valve. When a command from the cloud instructs the node to start irrigation, the ESP32 activates the relay; when the cloud sends a stop command, it de-energizes the relay, turning the water flow off. This ensures that every irrigation event is tied directly to cloud decisions rather than any local automatic logic.

A carefully designed power supply is crucial for reliable operation. Low-power parts of the system—sensors, ESP32, and LoRa module—normally run from a stable DC supply, such as 5 V or 3.3 V, provided by batteries, DC adapters, or solar-powered systems with charge controllers. Higher-power loads like pumps and some valves may require mains AC or higher DC voltages, so the design includes separate power rails, fuses, and breakers for safety. Voltage regulators and DC-DC converters are used to derive clean regulated voltages for the electronics, preventing brown-outs and erratic behaviour. Relay boards typically include opto-isolators or driver transistors to electrically separate the sensitive microcontroller circuitry from noisy high-voltage lines, reducing the risk of damage from surges or inductive spikes.

Because the system is intended for real farms, environmental protection and durability are built into the hardware design. Field nodes, gateways, and relay boxes are housed in weatherproof enclosures with appropriate IP ratings to protect against dust, rain, and irrigation spray. Cable glands and rubber seals are used at entry points to prevent water ingress, and UV-resistant materials are preferred for outdoor installations. Inside the enclosures, wiring is routed neatly using terminal blocks and secured to avoid loose connections due to vibration or

handling. Where necessary, conformal coating or protective varnish can be applied to PCBs to guard against moisture and corrosion.

Safety and reliability considerations also play a major role. Proper earthing and insulation are implemented wherever high-voltage AC is present, especially near pumps and mains wiring. Protective devices such as fuses, MCBs, and surge suppressors can be added to protect against faults, lightning strikes, or unstable rural grids. Clear separation is maintained between low-voltage signal wiring and high-voltage lines inside the enclosures to reduce the risk of accidental contact or interference. These design practices not only protect equipment but also improve user safety during installation and maintenance.

The overall hardware layout is modular and designed to scale. Each node is a repeatable building block: sensor interface, microcontroller, LoRa module, power supply, and optional relay outputs. If the farm area increases or new crop sections are added, more nodes can simply be deployed and registered with the system; the gateway and cloud architecture are already capable of handling multiple devices. Different types of sensors (temperature, humidity, light, or even flow meters) can be plugged into the same node design with minor firmware changes, making the platform flexible for broader smart-farm applications beyond just irrigation.

In essence, the hardware implementation brings the LoRaWAN-based, cloud-controlled concept into physical form. Controllers in the field do not “think” about irrigation; they sense conditions, communicate over long distances, and operate pumps and valves exactly as instructed by the cloud. By combining robust sensing hardware, reliable long-range communication, well-designed actuation interfaces, and protective mechanical and electrical design, the system delivers efficient water management with less manual effort and high dependability in real agricultural environments.

CHAPTER 10

RESULTS

The implemented smart irrigation system was evaluated under different soil and environmental conditions to verify how well it performs, how reliable the communication is, and whether the overall closed-loop control behaves as intended. Soil moisture sensors placed in the test plots were able to detect changes in soil water content clearly, showing distinct shifts between dry, moderately moist, and well-irrigated states. These readings were captured consistently by the field nodes and transmitted over LoRaWAN to the central gateway without noticeable data corruption or loss, confirming that the long-range communication layer is suitable for open agricultural environments.

The gateway successfully forwarded all collected data to the cloud platform using lightweight protocols such as MQTT, with only minor latency observed in normal conditions. Integration with the cloud backend (for example, AWS IoT Core) enabled live visualization of soil moisture trends, continuous logging of measurements, and server-side processing for decision-making. During testing, the system maintained a stable data stream with minimal delays between sensing in the field and updates shown on the dashboard, which is essential for timely irrigation control and remote supervision.

The decision-making capability of the system was validated using predefined soil moisture thresholds and simple rule-based logic. When soil moisture in a test zone fell below the configured lower threshold, the cloud logic recognized the dry condition and generated a command to start irrigation. This command traveled from the cloud to the gateway and then down to the relevant field node via LoRaWAN. On receiving the instruction, the node activated its relay, which in turn switched on the water pump or opened the solenoid valve feeding that zone. This behaviour confirmed that no local control logic was required at the node; all actuation actions were directly driven by cloud decisions, as intended in the architecture.

As irrigation proceeded, updated sensor data flowed back to the cloud, creating a live feedback loop. The system tracked how quickly the soil moisture rose and, once it reached or slightly exceeded the upper threshold, the cloud generated a “stop irrigation” command. This command was again routed through the gateway to the field node, which de-energized the

relay and stopped the water flow. In repeated tests, the system was able to stop irrigation shortly after the target moisture band was reached, avoiding prolonged over-watering and reducing the risk of waterlogging. This closed-loop behaviour aligns with findings from similar IoT-based irrigation systems, which show that feedback control significantly reduces water wastage compared with fixed-time watering.

When multiple field nodes were active at the same time, the system continued to perform reliably. Data from different locations were handled by a single gateway and processed centrally in the cloud without conflicts, demonstrating that the design scales to multiple zones or plots. Each node followed its own control commands based on the local sensor data associated with that node, while the cloud maintained a global view of all zones. This confirmed that the LoRaWAN-based communication and centralized decision-making can support larger fields or multi-crop layouts without needing separate standalone controllers in each area.

The incorporation of environmental context, such as weather information, further improved the quality of irrigation decisions. In scenarios where soil moisture was low but rainfall was forecast within a short period, the system could delay irrigation, thereby avoiding unnecessary water use and taking advantage of natural rainfall instead. This behaviour demonstrated the advantage of cloud-based processing, where external data sources can be easily integrated into the control logic. Compared with traditional systems that react only to local sensors or timers, this approach provides more intelligent and water-conscious irrigation scheduling.

From a resource and energy perspective, the field nodes operated at low power thanks to LoRaWAN's low-energy communication profile and duty-cycled measurements. This makes the setup practical for battery- or solar-powered deployments. At the same time, pumps and valves were active only when commanded, reducing unnecessary runtime and associated energy costs. Combined with reduced manual labour—since farmers no longer need to visit the field just to switch pumps—these factors contribute to tangible operational savings over time.

Overall, the test results indicate that the LoRaWAN-assisted cloud-controlled irrigation system functions as intended, with high accuracy in soil moisture control, reliable long-range communication, and robust closed-loop behaviour. The cloud platform consistently triggered irrigation based on analysed data, and all field actions followed these commands without

relying on local controllers for decision-making. The system achieved efficient water use, minimized manual intervention, and supported scalable coverage across multiple nodes, confirming that it meets its primary objectives and is well suited for modern smart farming applications.

Feature	Traditional Irrigation	Basic Automatic (Local Controller)	Proposed Cloud-Controlled System
Decision Making	Manual	Local controller	Cloud-based (centralized)
Real-Time Monitoring	No	Limited	Yes
Weather Integration	No	No	Yes
Water Efficiency	Low	Medium	High
Human Intervention	High	Medium	Low
Scalability	Poor	Limited	High
Accuracy	Low	Moderate	High

Table 10.1. Traditional vs Cloud-Controlled Irrigation

Parameter	Traditional Irrigation	Basic Automatic System	Proposed Cloud-Controlled System
Initial Setup Cost	Low	Moderate	Moderate (~₹2020)
Water Consumption Cost	High	Medium	Low
Labour Cost	High	Medium	Very Low
Maintenance Cost	Low	Moderate	Moderate
Energy Consumption	High	Medium	Low (optimized usage)
Operational Efficiency	Low	Medium	High
Long-Term Cost Savings	Low	Moderate	High

Table 10.1. Cost Comparison of Irrigation Systems

CHAPTER 11

ADVANTAGES

The proposed smart irrigation system offers several key advantages by combining IoT devices, long-range LoRaWAN communication, and centralized cloud intelligence into a single architecture. Unlike traditional or locally automated systems, all irrigation decisions are made in the cloud, while field nodes focus on sensing and executing commands. This separation of roles improves efficiency, accuracy, and reliability because the cloud always has a global view of conditions and can apply consistent rules across the entire farm.

A major advantage is efficient water management. Soil moisture is monitored continuously, and irrigation is triggered only when sensor data indicates that plants genuinely need water. This prevents both overwatering and underwatering, helping to keep the moisture level within an optimal band for crop growth. Studies on IoT-based irrigation have reported water savings in the range of 30–50% compared with traditional fixed-schedule systems, while maintaining or even improving yields, which highlights the potential of this approach for water-stressed regions.

The system also reduces human effort significantly. Because irrigation is driven by cloud-based automation, farmers no longer need to manually inspect soil conditions or physically operate pumps and valves as often. Instead, the entire process can be monitored and managed from a central dashboard or mobile app, which is particularly valuable for large or geographically dispersed farms. This reduction in routine labour lowers operational workload and allows farmers to focus more on crop management and planning.

Another important benefit is real-time monitoring and remote accessibility. By integrating with cloud platforms such as AWS IoT, users can view live soil moisture readings, system status, and irrigation history from anywhere with an internet connection. Alerts and notifications can also be configured for abnormal situations—for example, persistent dryness in a zone, device faults, or communication issues—enabling quicker responses to problems. This data-driven visibility improves decision-making and helps farmers learn how different parts of their fields behave over time.

The use of LoRaWAN communication adds further advantages. LoRaWAN provides long-range, low-power wireless connectivity, which means many field nodes can communicate with a single gateway over several kilometres without the need for complex wired infrastructure or expensive cellular subscriptions. This makes the system highly suitable for large-scale or remote farms where coverage and energy efficiency are critical. The same infrastructure can later support additional sensors and services as the operation grows, without major redesign.

Scalability and flexibility are also strong points. New nodes and sensors can be added easily, and different zones can be configured with their own thresholds and rules, making the system adaptable to various field sizes, crop types, and irrigation methods (drip, sprinkler, etc.). Because decision logic resides in the cloud, improvements or changes—such as new scheduling strategies or updated thresholds—can be rolled out centrally without touching field hardware. This makes the system easier to maintain and upgrade as requirements evolve.

In terms of accuracy and reliability, centralized analysis allows decisions to be based on real-time data and, when available, environmental inputs such as weather forecasts and temperature trends. The closed-loop feedback mechanism, where soil moisture is re-checked after irrigation and used to decide when to stop, ensures that water is applied precisely and only as long as needed. This leads to more consistent soil conditions and reduces the risk of both water stress and waterlogging.

These technical advantages translate into cost savings over time. Lower water usage directly reduces water bills or pumping costs, while shorter pump run times and optimized schedules help cut energy consumption. Cloud-based automation also reduces labour requirements, particularly on larger farms where manual irrigation is time-consuming and expensive. Although the initial setup involves hardware and configuration costs, the long-term savings and yield benefits can make the system economically attractive.

Finally, the system supports sustainable agriculture practices. By minimizing wasted water, reducing unnecessary energy use, and enabling more precise, data-driven irrigation, it contributes to better resource stewardship and environmental protection. The combination of IoT, LoRaWAN, and cloud computing represents a modern, future-ready approach to farming that helps transition from traditional methods towards smart, resilient, and climate-aware agricultural systems.

CHAPTER 12

FUTURE SCOPE

The proposed smart irrigation system, based on IoT, long-range communication, and cloud computing, has a wide range of applications in modern agriculture and water resource management. Since the system operates on a LoRaWAN-assisted cloud-controlled architecture, where field devices only collect data and execute commands while all decisions are made centrally in the cloud, it is particularly suitable for large-scale, distributed, and intelligent farming environments.

One of the most important applications of this system is in precision agriculture. In precision farming, maintaining optimal soil conditions is essential for maximizing crop productivity. The system continuously monitors soil moisture levels through distributed sensors and transmits this data using LoRaWAN technology to a cloud platform. The cloud processes this data along with additional inputs such as weather conditions and crop requirements to determine the exact irrigation needs. This ensures that water is supplied only when necessary, avoiding both over-irrigation and under-irrigation. As a result, crop health improves, water usage is optimized, and agricultural output becomes more consistent and efficient.

Another major application is in large-scale farming and commercial agriculture. Managing irrigation manually across vast agricultural lands is both time-consuming and inefficient. By deploying multiple field nodes across different zones of a farm, the system enables centralized monitoring and control of irrigation. All data is transmitted to the cloud and analysed collectively, allowing farmers to manage multiple sections of land from a single interface. This significantly reduces labour requirements and improves operational efficiency. Additionally, the long-range capability of LoRaWAN ensures reliable communication even in geographically dispersed fields, making it ideal for large farms.

Another critical application is in water-scarce and drought-prone regions. In areas where water availability is limited, efficient irrigation becomes essential. The system minimizes water wastage by ensuring that irrigation is carried out only when required, based on accurate soil moisture data and cloud-based analysis. The ability to integrate weather forecasts allows the system to avoid unnecessary watering, for example, by delaying irrigation when rainfall is expected. This contributes to sustainable water management and helps conserve valuable water resources, making the system highly suitable for regions facing water shortages.

Another important application is in remote farm monitoring and management. Since the system is integrated with a cloud platform such as Amazon Web Services, users can access real-time data and system status from anywhere through web or mobile interfaces. This is particularly useful for farmers who manage multiple farms or cannot be physically present at all times. They can monitor soil moisture levels, track irrigation activity, and even manually override the system if necessary. This level of accessibility improves decision-making and reduces dependency on on-site labour.

In addition, the system can be utilized in smart city and urban agriculture initiatives. With the growing trend of urban farming, rooftop gardens, and vertical farming, efficient water management becomes crucial. The proposed cloud-controlled irrigation system can be integrated into such setups to automate watering and optimize resource usage. Its scalability and remote monitoring capabilities make it suitable for integration into larger smart city infrastructures.

Another emerging application is in plant nurseries and horticulture industries, where maintaining consistent soil moisture is essential for plant growth and quality. The system ensures uniform watering across different sections of the nursery, improving plant health and reducing losses. Since the system is scalable, it can be expanded as the nursery grows, making it a practical long-term solution.

Furthermore, the system supports integration with advanced IoT and data analytics platforms. By using communication protocols such as MQTT, the system can easily connect with dashboards, analytics tools, and machine learning models. This opens up possibilities for predictive irrigation, crop-specific watering strategies, and automated decision-making based on historical data. Such advanced applications further enhance the efficiency and intelligence of the irrigation process.

Overall, the applications of the LoRaWAN-assisted cloud-controlled irrigation system extend across multiple domains, from small-scale gardening to large-scale commercial farming. Its ability to combine real-time sensing, long-range communication, and centralized cloud intelligence makes it a versatile and powerful solution for modern agriculture. By reducing water wastage, minimizing manual effort, and enabling data-driven decision-making, the system plays a crucial role in transforming traditional irrigation practices into smart, efficient, and sustainable solutions.

CHAPTER 13

CONCLUSION

The IoT-based smart irrigation system developed in this project presents an advanced approach to modernizing traditional agricultural practices through the integration of sensing, long-range communication, and cloud-based intelligence. Unlike conventional automated systems, this project emphasizes a LoRaWAN-assisted, cloud-controlled architecture, where field devices are used only for data collection and execution, while all irrigation decisions are made centrally in the cloud. This ensures more accurate, flexible, and data-driven control of irrigation processes.

The system successfully demonstrates continuous monitoring of soil moisture conditions using distributed field nodes. These nodes transmit real-time data over long distances using LoRaWAN technology to a central gateway, which then forwards the information to the cloud. The cloud platform processes this data, integrates additional inputs such as environmental and weather conditions, and determines whether irrigation is required. Based on this analysis, control commands are sent back to the field, ensuring that water is supplied only when necessary. This confirms that no autonomous decision-making is performed at the controller level, making the system more intelligent and adaptable.

One of the key achievements of the project is the reliable implementation of long-range communication combined with cloud integration. The use of LoRaWAN enables efficient and low-power communication across large agricultural areas, making the system suitable for rural and remote environments. The integration with cloud platforms, such as Amazon Web Services, along with communication protocols like MQTT, allows real-time monitoring, secure data transmission, and centralized control. This significantly enhances the overall performance and usability of the system.

The project also proves to be cost-effective, with an approximate implementation cost of ₹2020, making it accessible for small and medium-scale farmers. Despite its affordability, the system provides substantial benefits, including reduced manual labour, improved water management, and enhanced operational efficiency. By automating irrigation through cloud-based decision-making, the system eliminates the need for constant human supervision and minimizes the risks associated with improper watering practices.

In addition, the system contributes to sustainable agriculture by optimizing water usage and reducing resource wastage. The ability to incorporate weather data into the decision-making process further improves efficiency by preventing unnecessary irrigation. Maintaining optimal soil moisture levels not only conserves water but also supports healthier plant growth and potentially increases crop yield.

Overall, the project successfully demonstrates how the integration of IoT, cloud computing, and wireless communication can transform conventional irrigation systems into intelligent and scalable solutions. The LoRaWAN-assisted cloud-controlled model provides a strong foundation for future enhancements such as predictive analytics, AI-based decision-making, and fully integrated smart farming systems. This work represents a significant step toward achieving efficient, sustainable, and technology-driven agriculture for the future.

Furthermore, the system highlights the importance of shifting from localized control mechanisms to centralized, cloud-driven intelligence in modern agricultural applications. By separating sensing, communication, and decision-making processes, the system achieves greater flexibility and adaptability to varying environmental conditions. This architecture not only improves current irrigation practices but also enables seamless integration with emerging technologies such as advanced data analytics, remote sensing, and automated farm management systems. Such integration can further enhance decision accuracy and resource optimization.

Finally, the project opens up opportunities for large-scale deployment and real-world implementation in diverse agricultural scenarios. With further improvements in hardware durability, network reliability, and user-friendly interfaces, the system can be adopted widely by farmers and agricultural organizations. The combination of affordability, scalability, and intelligent control makes this solution a promising step toward the future of smart agriculture, where efficient resource utilization and sustainable practices are essential for meeting growing global food demands.

CHAPTER 14

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